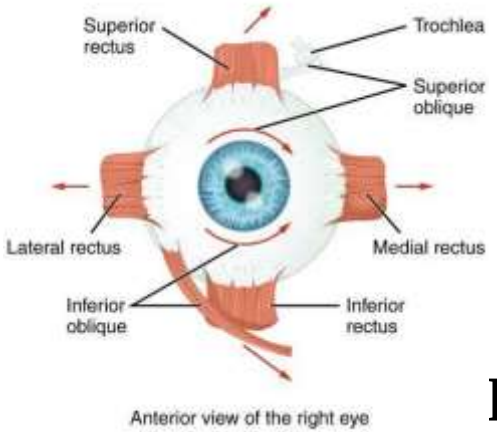
BASIC NEUROLOGY- CRANIAL NERVES III, IV, VI



Anterior view of the right eye

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Introduction

- The 3rd, 4th and 6th cranial nerves are pure motor nerves which are related to extra ocular movements. The 3 nerves are so closely related that they are considered one single system- The oculomotor system.

- The 3rd nerve supplies the sphincter pupillae (autonomic supply), LPS, IR, IO, SR and the MR.

- The 4th nerve supplies the SO and the 6th nerve supplies the LR .

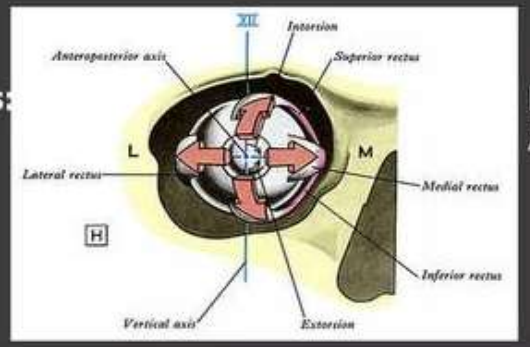
Extraocular muscles and functions

MUSCLE	PRIMARY ACTION	SECONDARY ACTION	TERTIARY ACTION
MEDIAL RECTUS	ADDUCTION	-	-
LATERAL RECTUS	ABDUCTION	-	-
SUPERIOR RECTUS	ELEVATION	INTORTION	ADDUCTION
INFERIOR RECTUS	DEPRESSION	EXTORTION	ADDUCTION
SUPERIOR OBLIQUE	INTORTION	DEPRESSION	ABDUCTION
INFERIOR OBLIQUE	EXTORTION	ELEVATION	ABDUCTION

Extraocular muscles and functions

Superior rectus:
Elevation; Intortion; Adduction

Lateral rectus:
Abduction

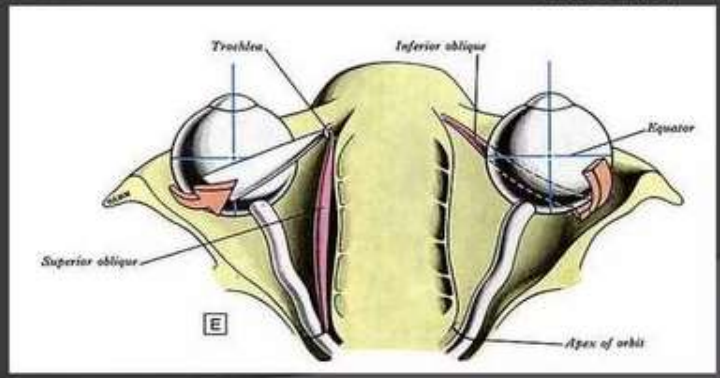


Medial rectus:
Adduction;

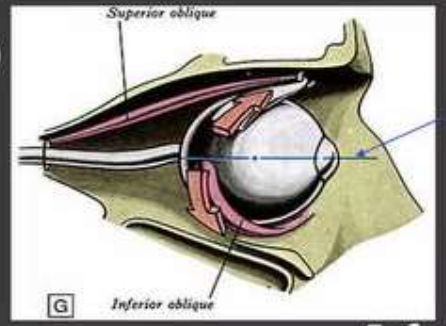
Inferior rectus:
Depression; Extortion; Adduction

Superior Oblique:
Intortion,
Depression,
Abduction

Inferior Oblique:
Extortion,
Elevation,
Abduction



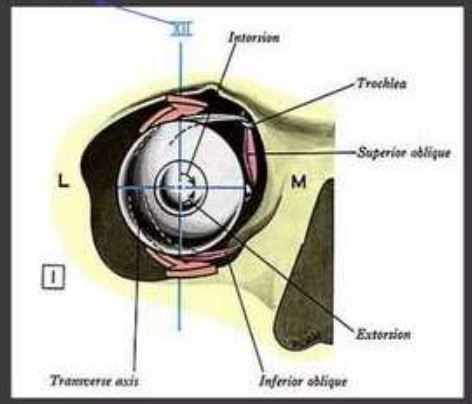
Superior Oblique:
Intortion
(Anterior Fibers)



Anteroposterior axis

Inferior Oblique :
Extortion

Vertical axis



Both oblique muscles pulls posterolateral quadrant anteromedially; thus abduct the eyeball.

Examination of 3rd, 4th, 6th Cranial nerves

- Inspection: Ptosis, strabismus (U/L or B/L) (constant or variable), exophthalmos or enophthalmos, hyper/hypotelorism, lid retraction.
- Pupils: size, shape, reactivity to direct and indirect light reflex, anisocoria.
- Accommodation reflex.
- Nystagmus.
- Extraocular movements of individual muscles as described.

Clinical actions of extraocular muscles

- LR: Abduction
- MR: Adduction
- SR: Elevation in an abducted eye
- IR: Depression in an abducted eye
- SO: Depression in an adducted eye
- IO: Elevation in an adducted eye

OCULOMOTOR NERVE

COURSE OF 3RD NERVE

Nucleus in upper midbrain

↓

Axons pass through red nucleus, substantia nigra, cerebral peduncles

↓

Nerve exits midbrain through interpeduncular fossa

↓

Enters subarachnoid space

↓

Pierces the dura

↓

Travels through the lateral wall of the cavernous sinus

↓

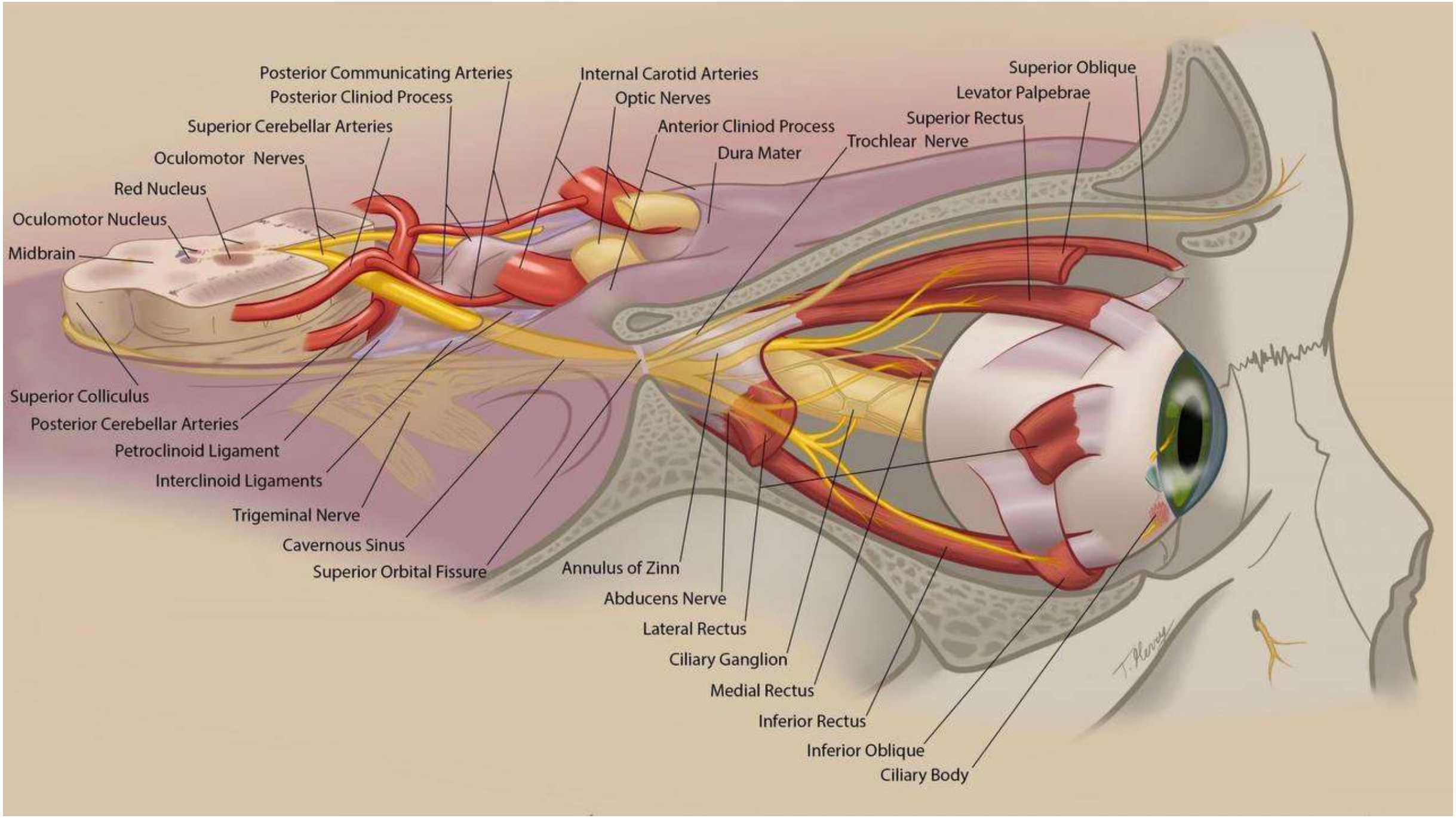
Passes through the middle part of superior orbital fissure

↓

Enters the Orbit

↓

Fibres divide to supply individual muscles and the parasympathetic fibres go to the ciliary ganglion to supply the pupil and ciliary muscles


Posterior Communicating Arteries
Posterior Cliniod Process
Superior Cerebellar Arteries
Oculomotor Nerves
Red Nucleus
Oculomotor Nucleus
Midbrain
Internal Carotid Arteries
Optic Nerves
Anterior Cliniod Process
Dura Mater
Trochlear Nerve
Superior Oblique
Levator Palpebrae
Superior Rectus
Superior Colliculus
Posterior Cerebellar Arteries
Petroclinoid Ligament
Interclinoid Ligaments
Trigeminal Nerve
Cavernous Sinus
Superior Orbital Fissure
Annulus of Zinn
Abducens Nerve
Lateral Rectus
Ciliary Ganglion
Medial Rectus
Inferior Rectus
Inferior Oblique
Ciliary Body

CRANIAL NERVE III- Oculomotor Nerve Anatomy

- Origin: Nucleus located in Upper midbrain at level of Superior colliculus.

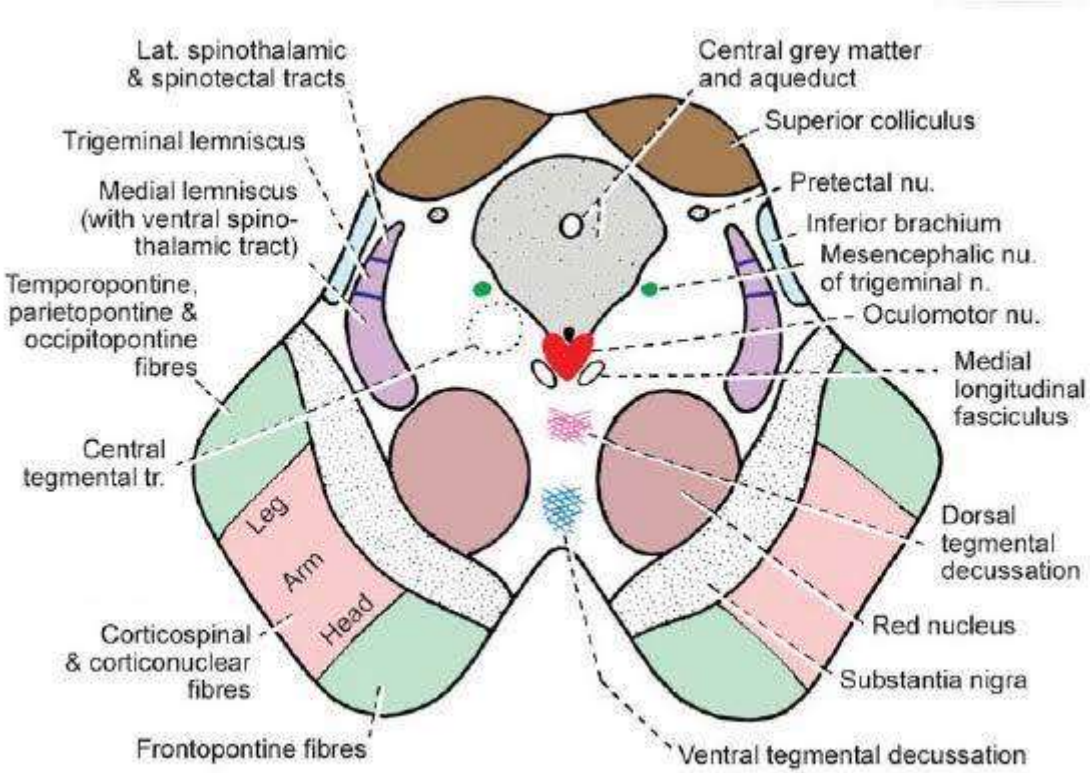


Fig. 11.9. Transverse section through the upper part of the midbrain.

Oculomotor nerve nucleus

- Located vertically in form of 5mm Rostrocaudal columns

- Has 1 unpaired core (central nucleus) with 4 paired subnucleus. (total of 9 subnucleus)

- **CENTRAL NUCLEUS** Upper part has Edinger Westpal Nucleus which controls the parasympathetic part of pupil
- **CENTRAL NUCLEUS** Lower part has nucleus for Levator palpebrae superioris muscle for lid elevation

Oculomotor Nerve nucleus- Paired Subnucleus

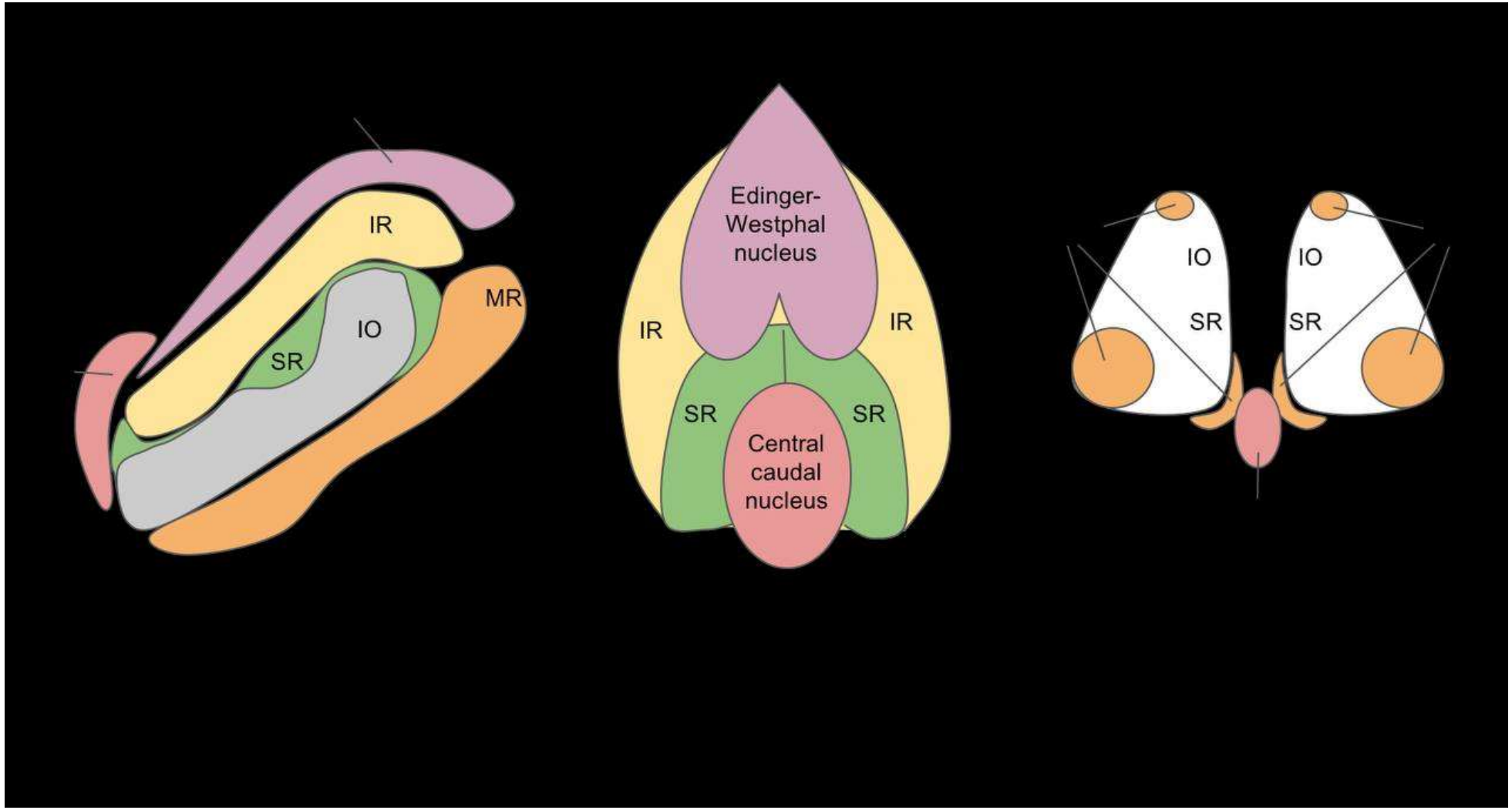
- **SUPERIOR RECTUS SUBNUCLEUS:** Fibers from this nucleus cross to opposite side within the contralateral nucleus to supply opposite side superior rectus.

- **MEDIAL RECTUS SUBNUCLEUS**

- **INFERIOR OBLIQUE SUBNUCLEUS**

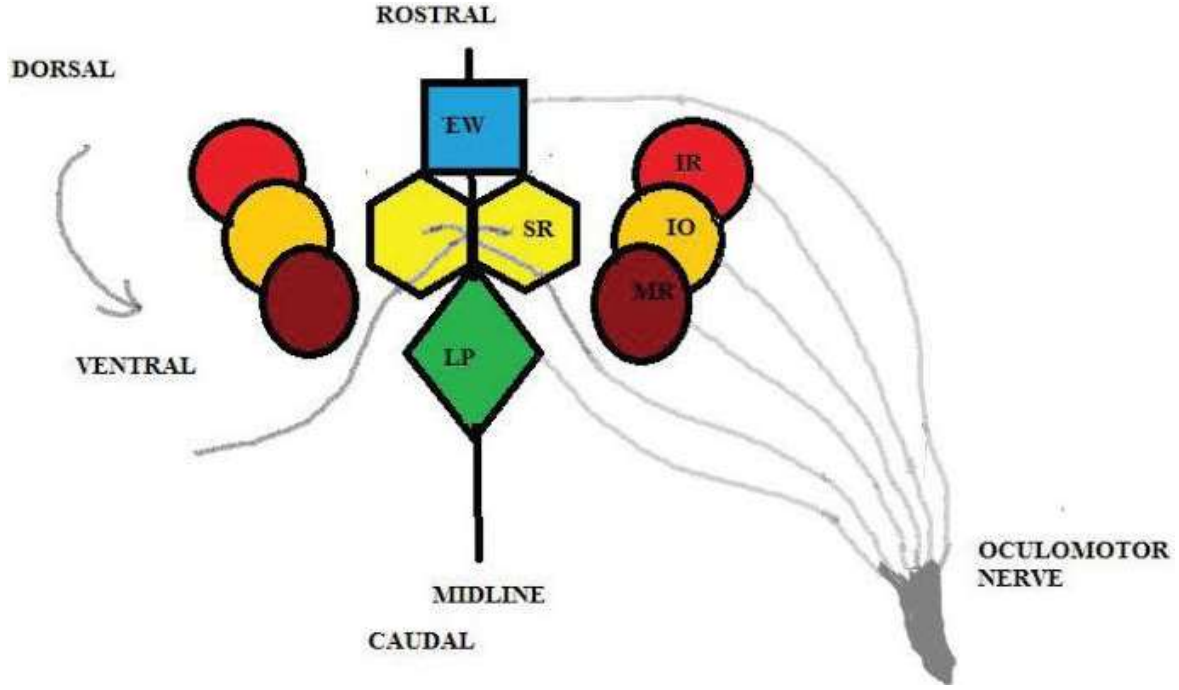
- **INFERIOR RECTUS SUBNUCLEUS**

Nuclear structure of 3rd nerve



Nuclear III Nerve Palsy

- B/L pupillary involvement (as EWN is present in the central subnucleus- supplies bilateral pupils)
- B/L Ptosis (as bilateral LPS is supplied by central subnucleus)
- B/L superior rectus palsy



Fascicular Organisation and significance

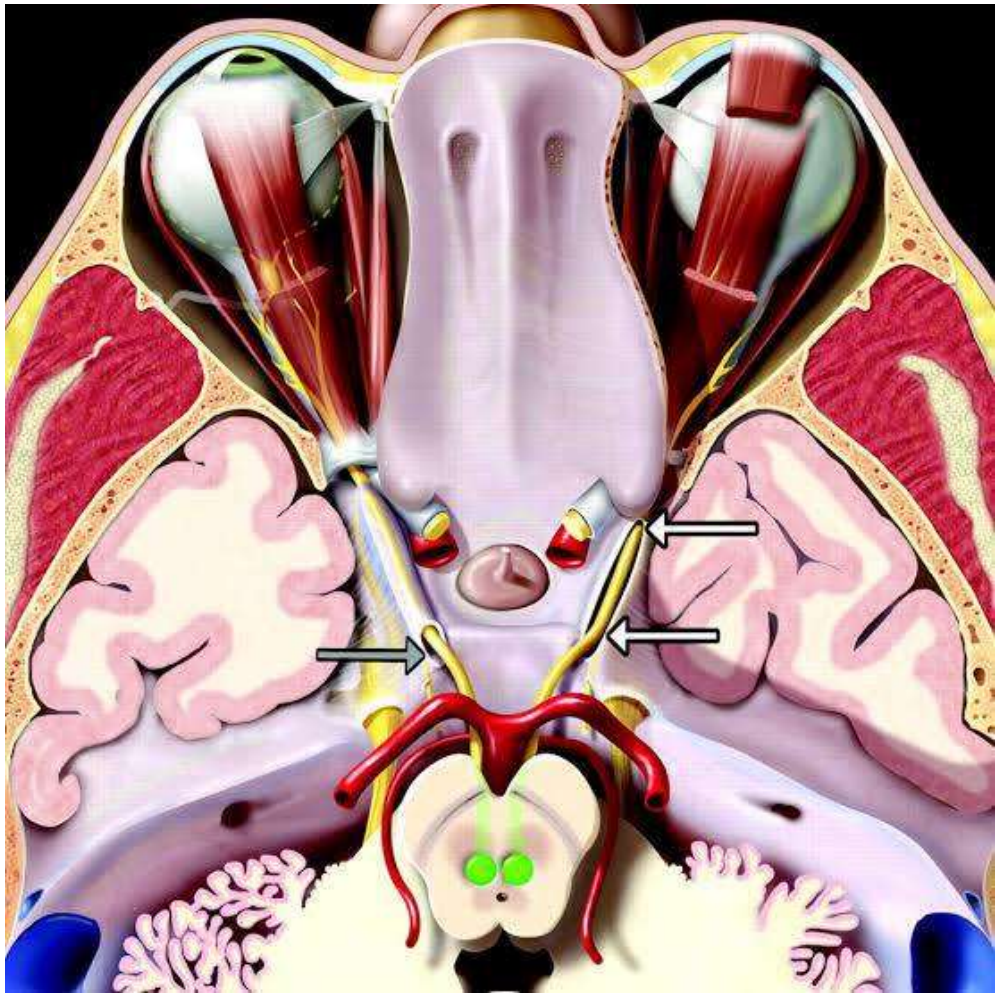
- Fibers of pupil are located laterally and fibres of inferior oblique are located medially. Hence paramedian lesions of midbrain can spare pupils.

- Vertically, fibres of SR and LPS travel in lower part of the fascicle (which later form superior division of 3rd nerve), and fibres of pupils, IR, IO, MR travel in the upper part of the fascicle (which later form inferior division of 3rd nerve).

- Hence lesion of lower part of the brainstem column can resemble isolated superior division palsy.

Interpeduncular Fossa

- Here the nerve passes between SCA and PCA. Hence any aneurysm of these vessels can compress the medial and inferior part of the nerve.



Interpeduncular Fossa

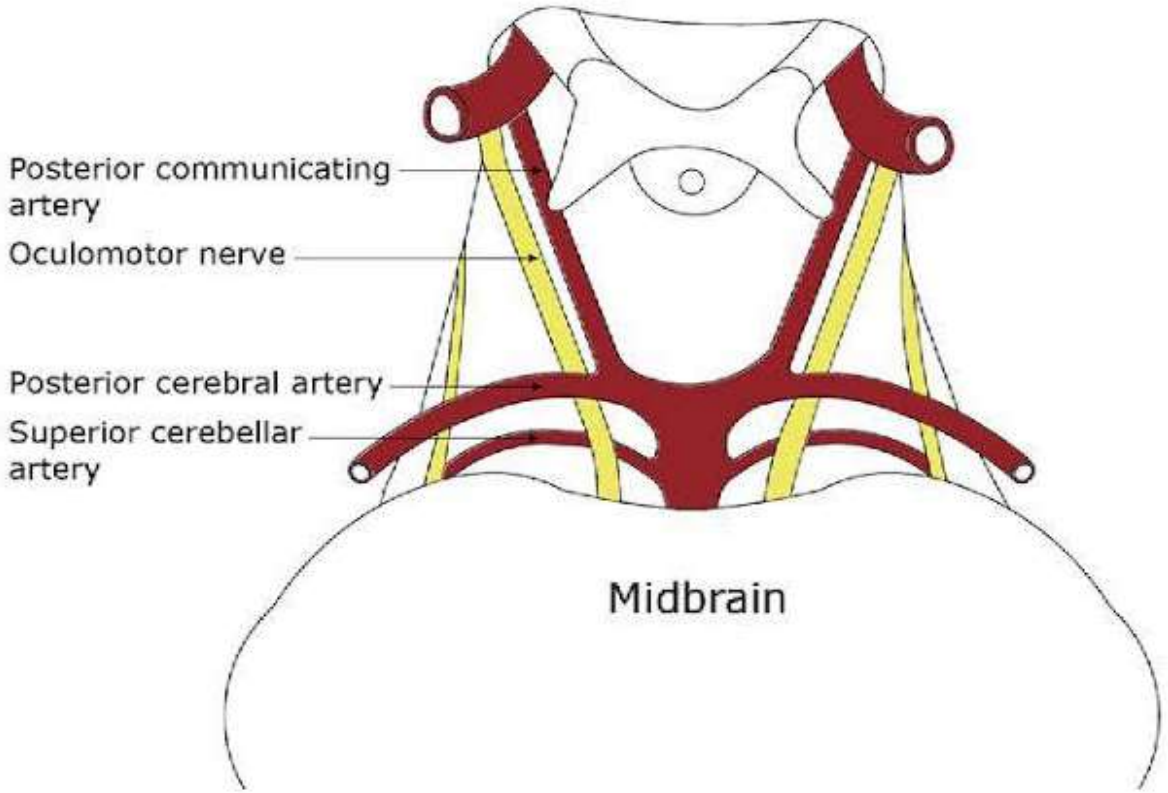
- Pupillary fibres are arranged peripherally. Hence aneurysms usually affects the pupils.

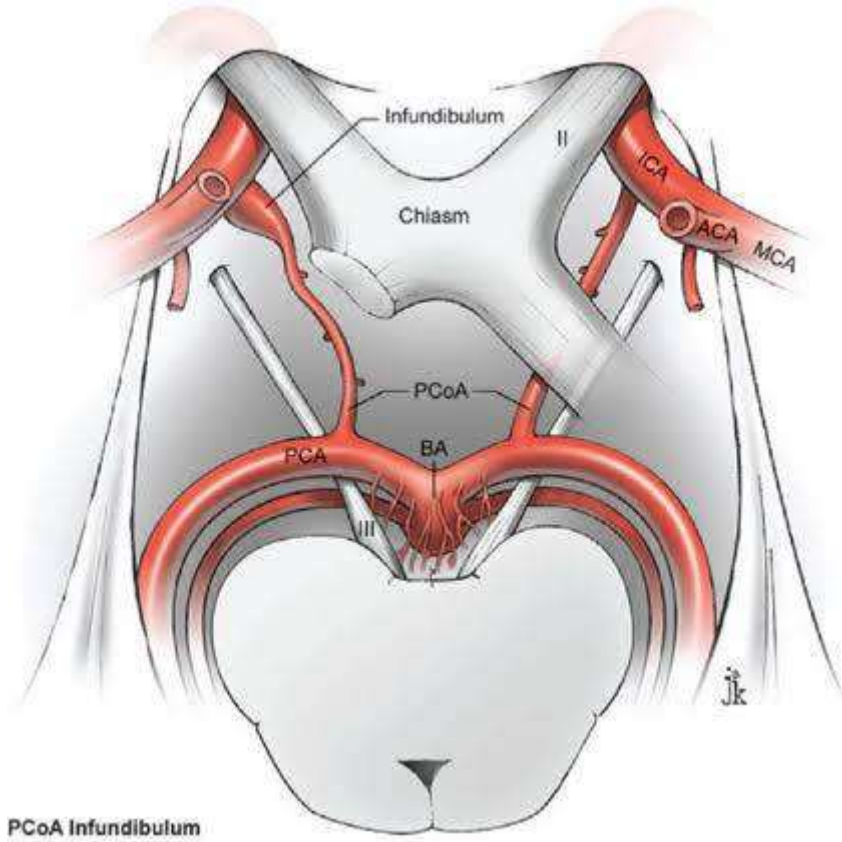
- PCA/SCA aneurysms usually present with involvement of pupil and SR+LPS (fibres present in lower part of the nerve)

- Diabetic neuropathy/ Ischemic neuropathy- Here central fibres are affected more (as blood supply is from periphery to centre), and hence there is sparing of pupils usually.

Subarachnoid Space

- After passing through the interpeduncular fossa, the nerve enters the subarachnoid space.
- Here it lies in close proximity to the PCOM artery parallelly.
- Hence PCOM aneurysms also present with 3rd nerve palsies, however here pupillary involvement is seen while SR+LPS involvement is rare.
- It also lies in close proximity to the uncus of temporal lobe here and hence uncal herniation can cause a Hutchinson pupil.

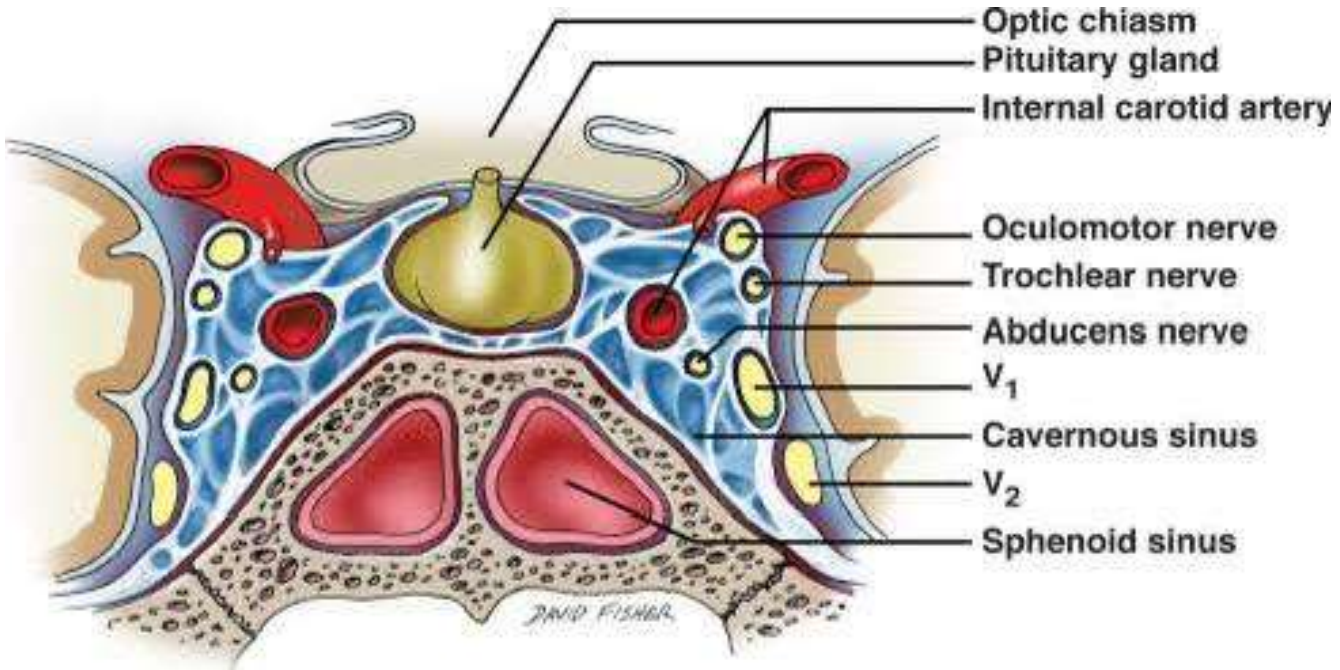




PCoA Infundibulum

Cavernous Sinus

- After passing through the subarachnoid space, the nerve pierces the dura to enter the cavernous sinus.
- The relations are as follows:



Lesions related to cavernous sinus

- Posterior cavernous involvement: V1 + V2 + 3 + 4 + 6 + Sympathetic nerves involved, sometimes V3 can be involved due to its close proximity to posterior part of cavernous sinus.

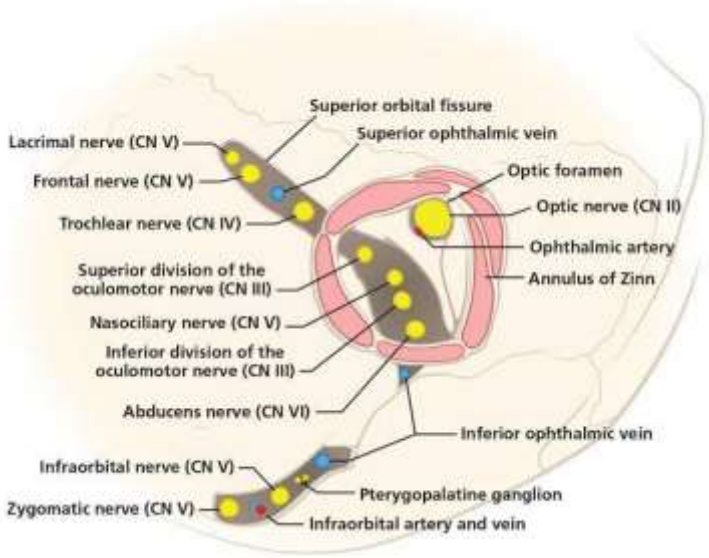
- Proper/ Middle cavernous involvement: V1 + V2 + 3 + 4 + 6 + Sympathetic nerves involved.

- Anterior cavernous involvement: V1 + 3 + 4 + 6, V2 may or may not be involved as it exits the sinus in the middle part

Superior Orbital Fissure

- The nerve just anterior to the cavernous sinus divides in to superior division (contains fibres of lower fasciculus- pupil + LPS), and inferior division (contains fibres from upper fasciculus- pupils, IR, IO, MR).

- Both the divisions enter the orbit through the middle part of the superior orbital fissure along with nasociliary nerve and 6th nerve.



Orbital Apex

- After passing through the superior orbital fissure, the nerve enters the orbit and innervates the extraocular muscles and supplies ciliary ganglion (preganglionic parasympathetic supply).

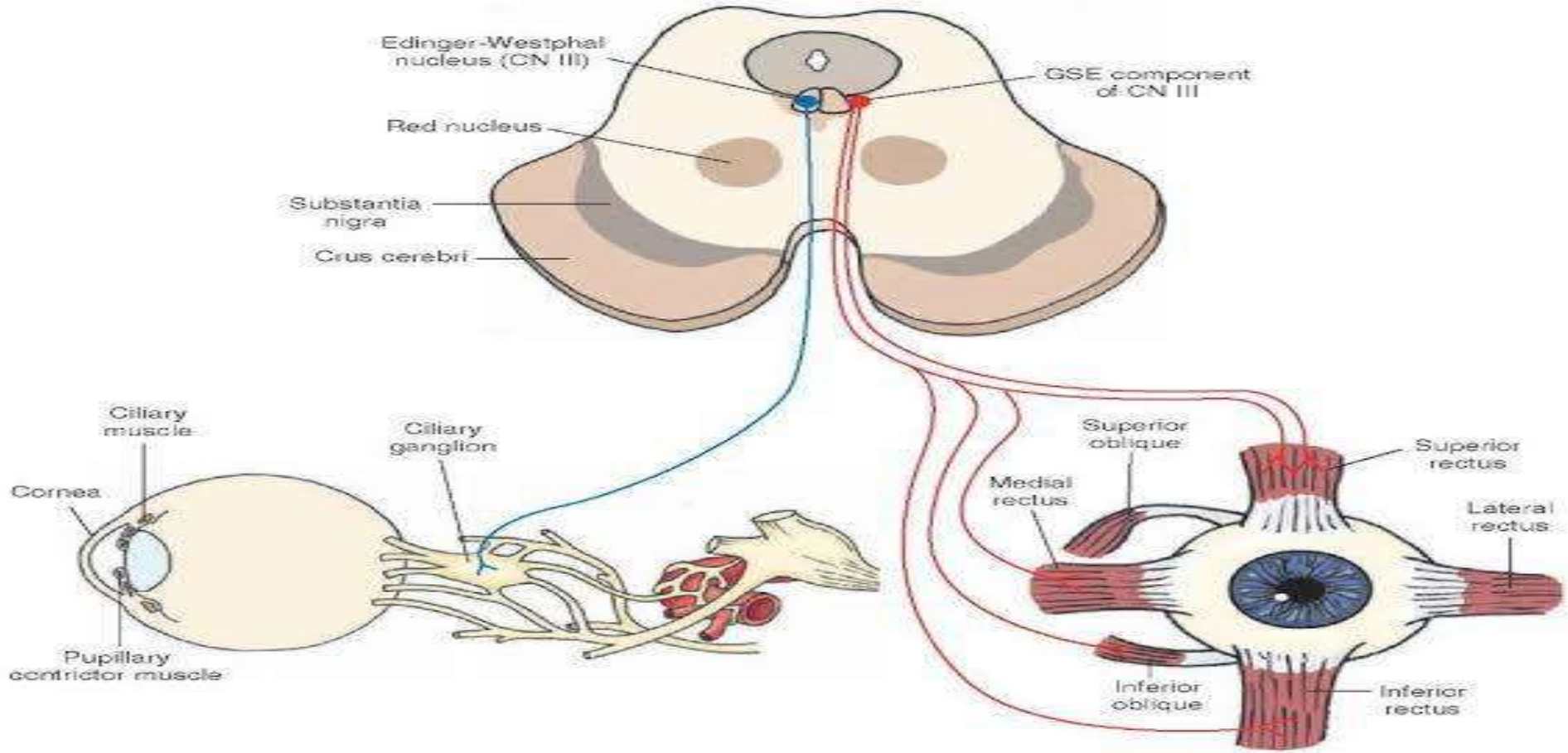
- Lesions of 3rd nerve at the orbital apex are associated with proptosis with flaccid eyeball, separate superior or/and inferior division involvement and optic nerve involvement.

Oculomotor nerve functions

- Motor/ Somatic functions: It innervates the SR, IR, MR, IO and the LPS muscle. These muscles are responsible for extraocular movements.

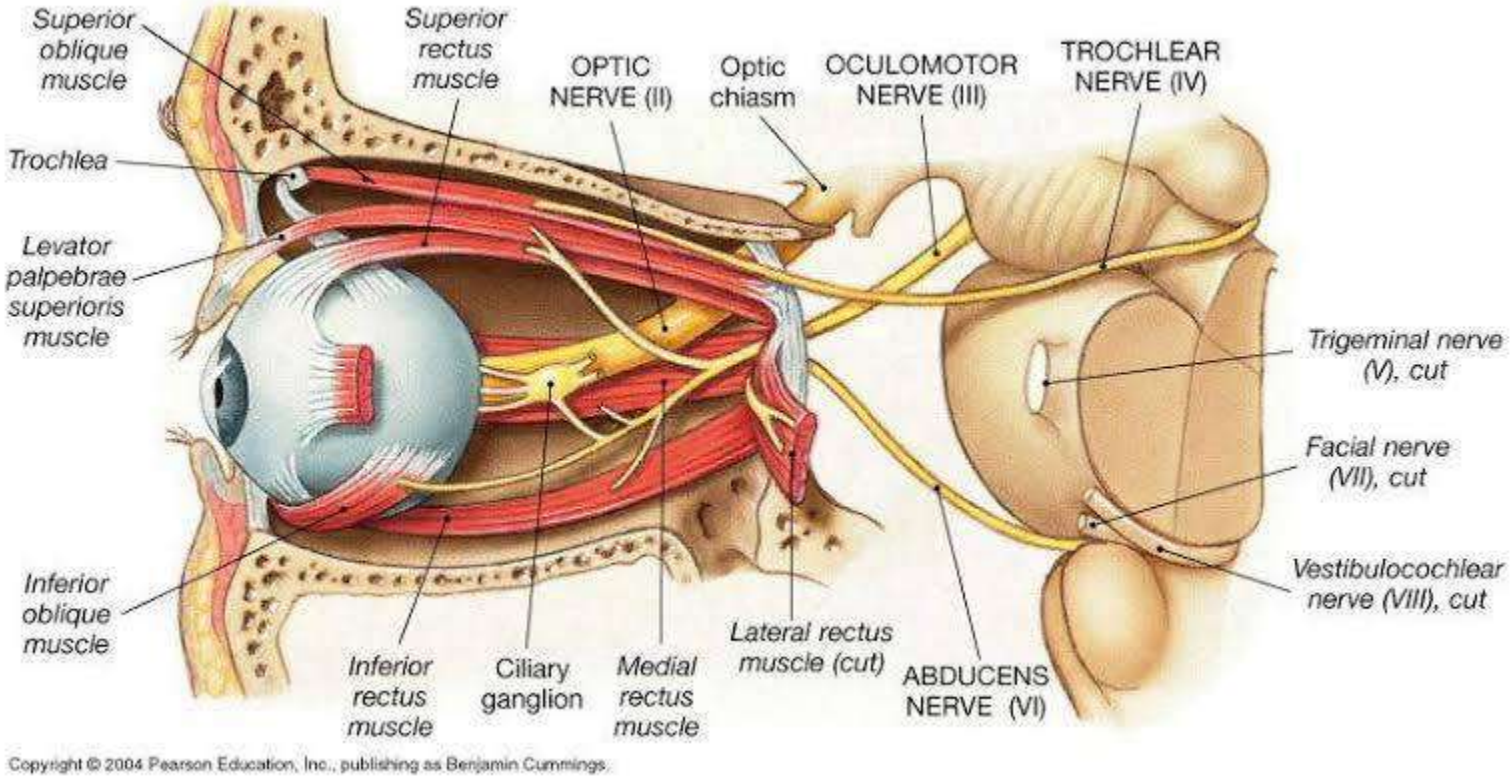
- Autonomic functions: The nerve provides parasympathetic innervation to the sphincter pupillae (for pupillary constriction) and the ciliary muscles (for lens accommodation) via the ciliary ganglion.

Functions of 3rd nerve



Parasympathetic supply

- Parasympathetic fibres from superior division of oculomotor nerve gives fibres to ciliary ganglion where it relays.
- After relaying in the ganglion, it exits via 8-10 short ciliary nerves supplying the sphincter pupillae and ciliary muscles.

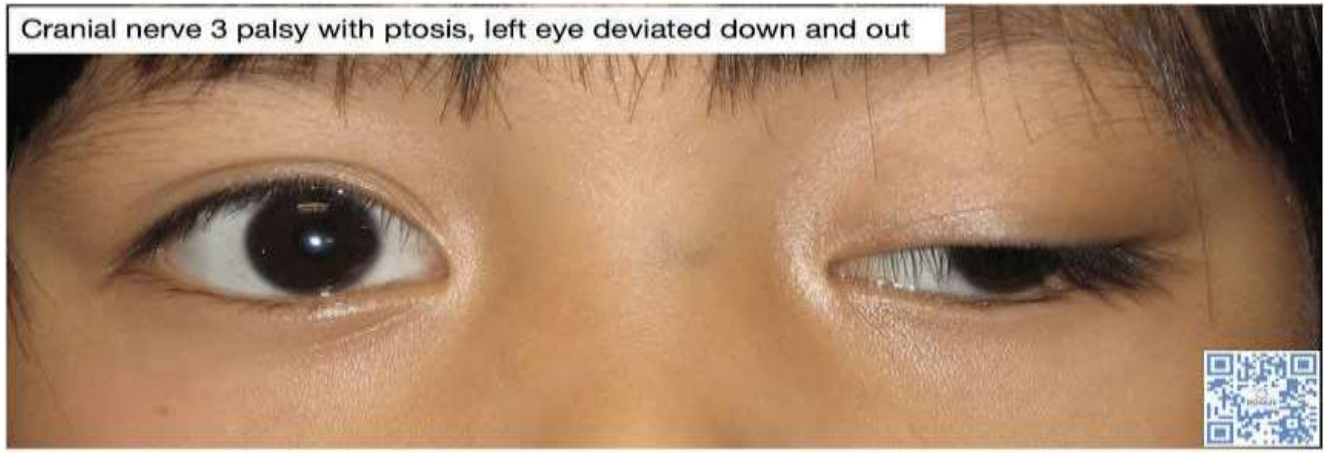


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Causes of oculomotor nerve palsy

- Vascular disorders: Ischemia due to diabetes, atherosclerosis, etc
- Midbrain stroke syndromes (Weber's, Benedikts syndrome)- nuclear/fascicular palsy
- Trauma including head injuries
- Infections: meningitis, encephalitis
- Neoplastic: Mass lesions compressing the nerve
- Aneurysms: Involving SCA, PCA, PCOM
- Congenital factors: Rare developmental anomalies
- Iatrogenic causes: Surgeries around brainstem/ orbit

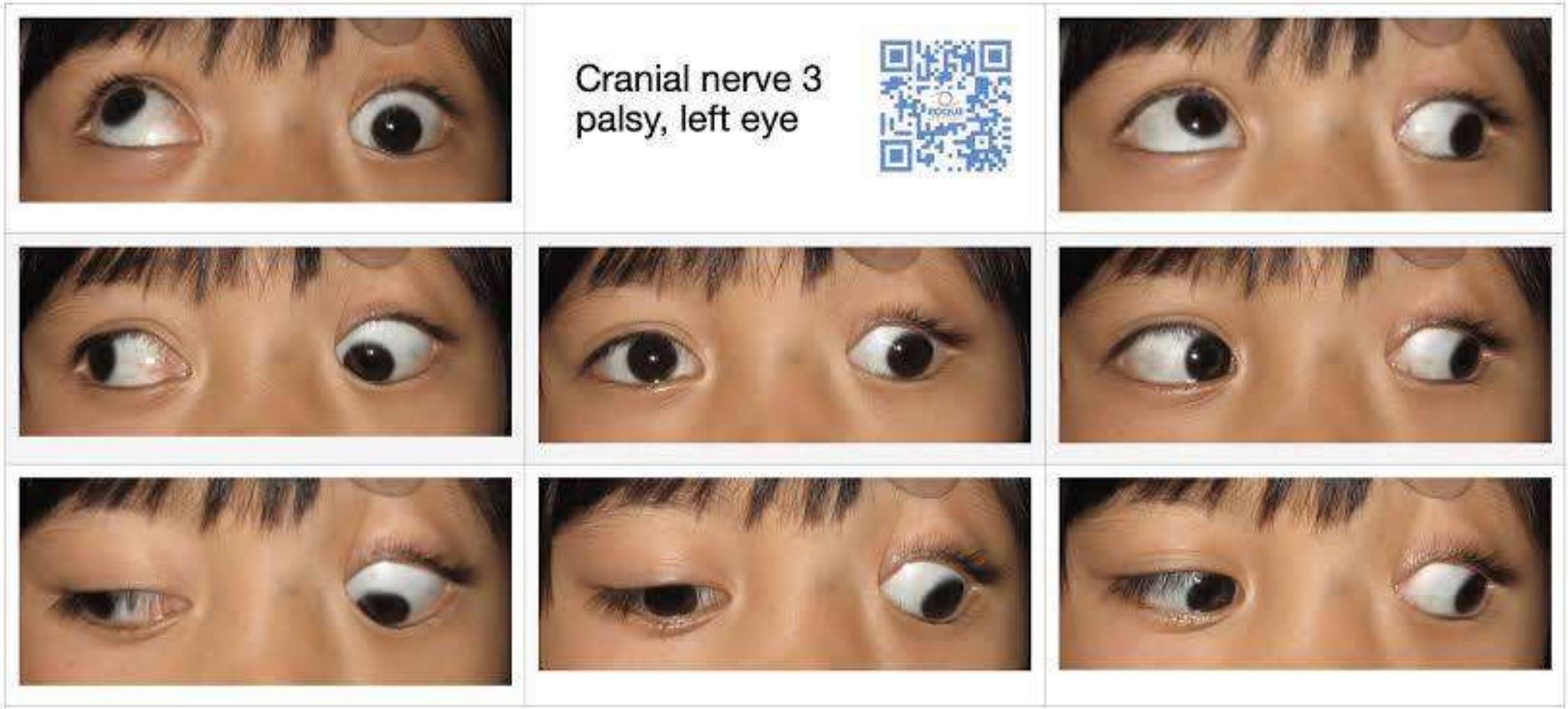
Clinical Features



Injury to the 3rd nerve can cause

- Double vision (diplopia) due to extraocular muscle palsy
- Primary gaze: eye turned downward, outward and intorted
- Impaired upward, downward and medial gaze
- Squint (strabismus)
- Ptosis (drooping eyelids)
- Mydriasis of pupil, not reacting to direct and indirect light reflex
- Difficulty in near vision.

Secondary gaze in 3rd nerve palsy



Diagnostic approaches

- Clinical examination: lid position, pupil size, extraocular movements.
- Imaging: MRI or CT to identify aneurysms (angiography), tumours, other structural causes.
- Blood investigations: for diabetes, infections, or inflammatory lesions.
- CSF examinations: if an infection of the CNS is suspected.

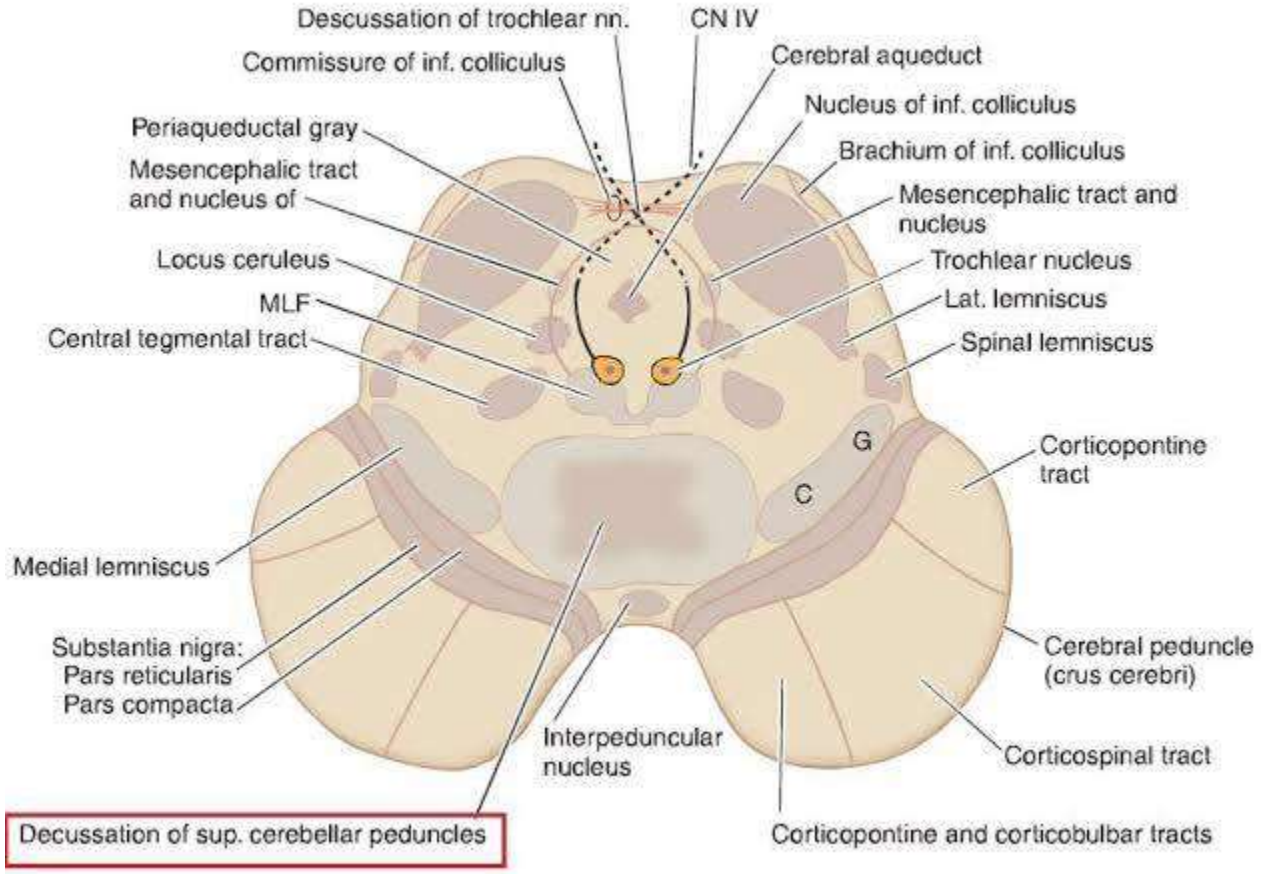
Treatment Options

- Addressing underlying cause: diabetes, hypertension, etc.
- Surgical interventions: for tumours or aneurysms causing nerve compression.
- Corticosteroids: if inflammation is suspected.
- Eye protection: Artificial tears.
- Prism glasses and eye patching: to relieve diplopia and prevent amblyopia

TROCHLEAR NERVE

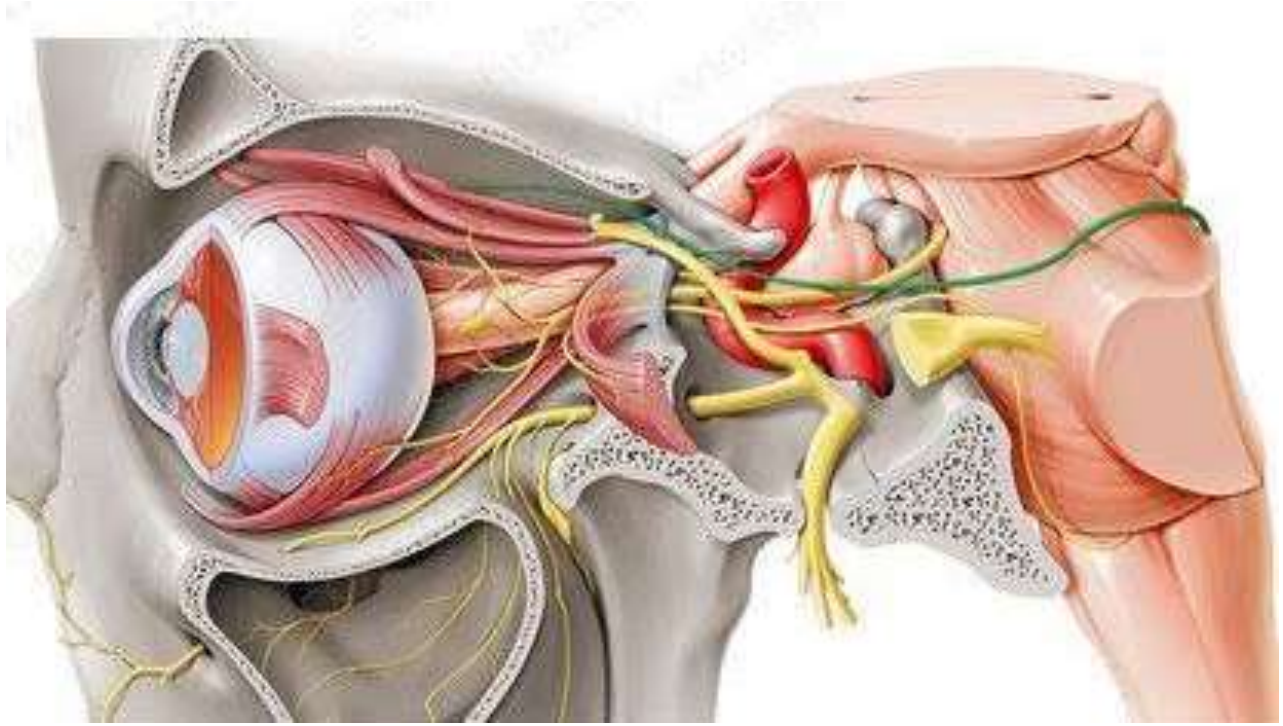
Nerve Nucleus

- Present in midbrain at the level of inferior colliculus.
- The nerve fibres crosses to the opposite side inside the parenchyma.
- The nerve emerges posteriorly.



Course of the nerve

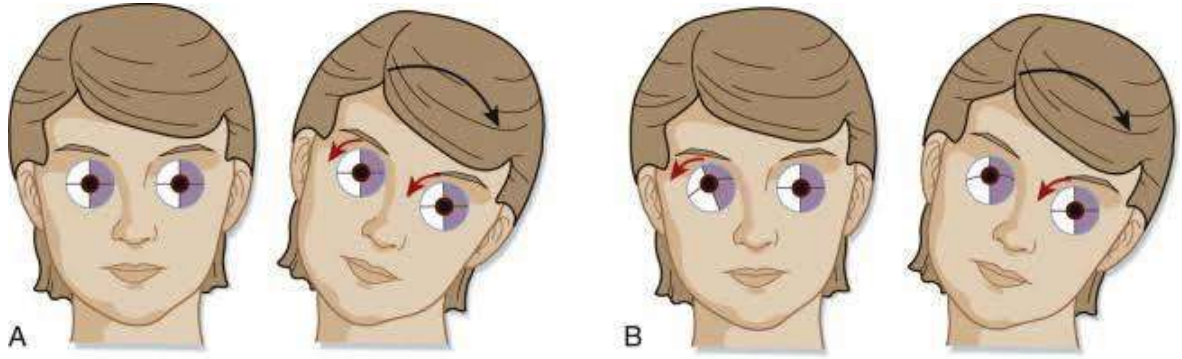
- Nucleus → Fibres crosses to opposite side inside the parenchyma→ arises posteriorly→ rounds around midbrain in peri mesencephalic area and comes anteriorly→ Pierces the dura and enters into the cavernous sinus→ exits through superior orbital fissure in the upper part→ enters orbit to supply superior oblique muscle.



Function and significance

- The nerve innervates the superior oblique muscle causing depression and abduction of the eyeball.

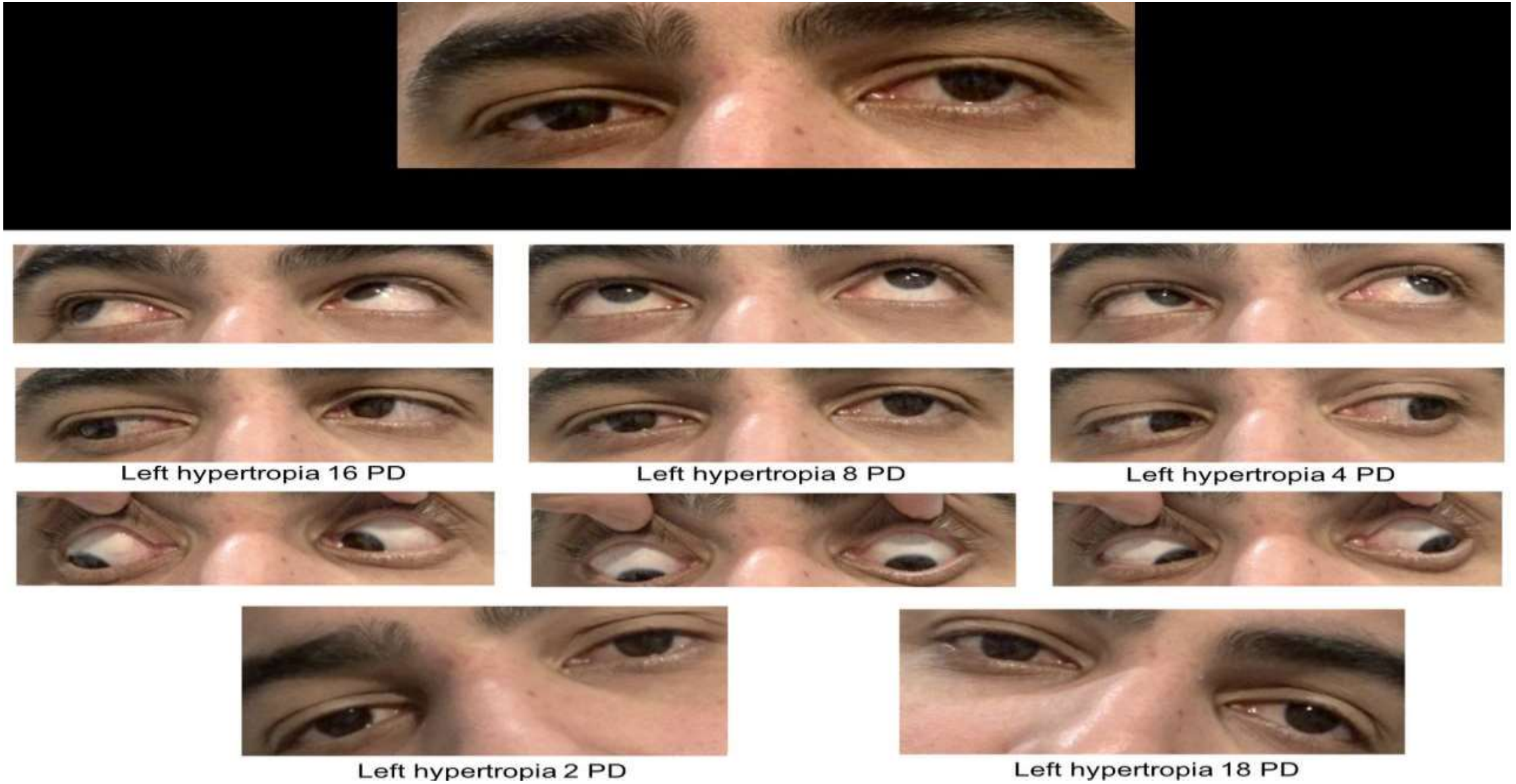
- Lesion in the trochlear nerve causes vertical diplopia (images stacking upon each other) especially when looking down, causing head tilt to the opposite side. [DOWNGAZE PALSY IN AN ADDUCTED EYE].



Normal eye rotation
When the head tilts to the left, both eyes rotate in the opposite direction (right eye extorts, left eye intorts)

Cranial nerve IV palsy (right eye)
Right eye extorted and slightly elevated causing double vision. To compensate, the patient tilts her head to the left

Primary and Secondary gaze



Causes of 4th nerve lesion

- Barotrauma (ascent and descent of flight)- as this nerve is the smallest cranial nerve.
- Microvascular diseases like diabetes.
- Trauma.
- Infections as in meningitis or meningoencephalitis.
- Tumors, aneurysms causing compressions.
- Neurodegenerative disorders.
- Congenital/ Developmental anomalies.

Management

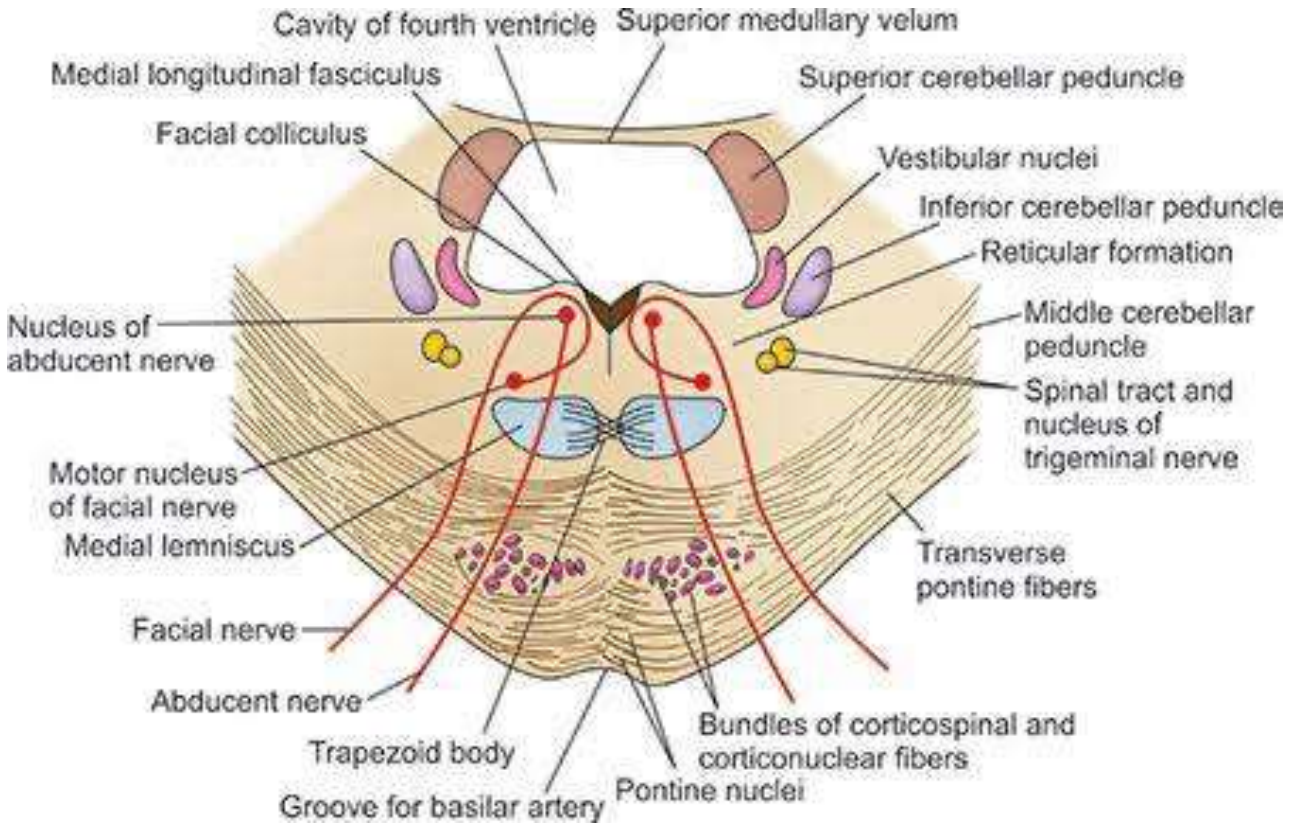
- Diagnostic methods: Clinical and neurological examination, Imaging (CT/MRI for mass lesions, Angiography for aneurysms, etc).

- Treatment strategies: Observation, Prismatic lenses for diplopia, Eye exercises and eye muscle surgeries are the options

ABDUCENS NERVE

Course and Anatomy

- Nucleus: In mid pons, at floor of 4th ventricle at the level of facial colliculus.

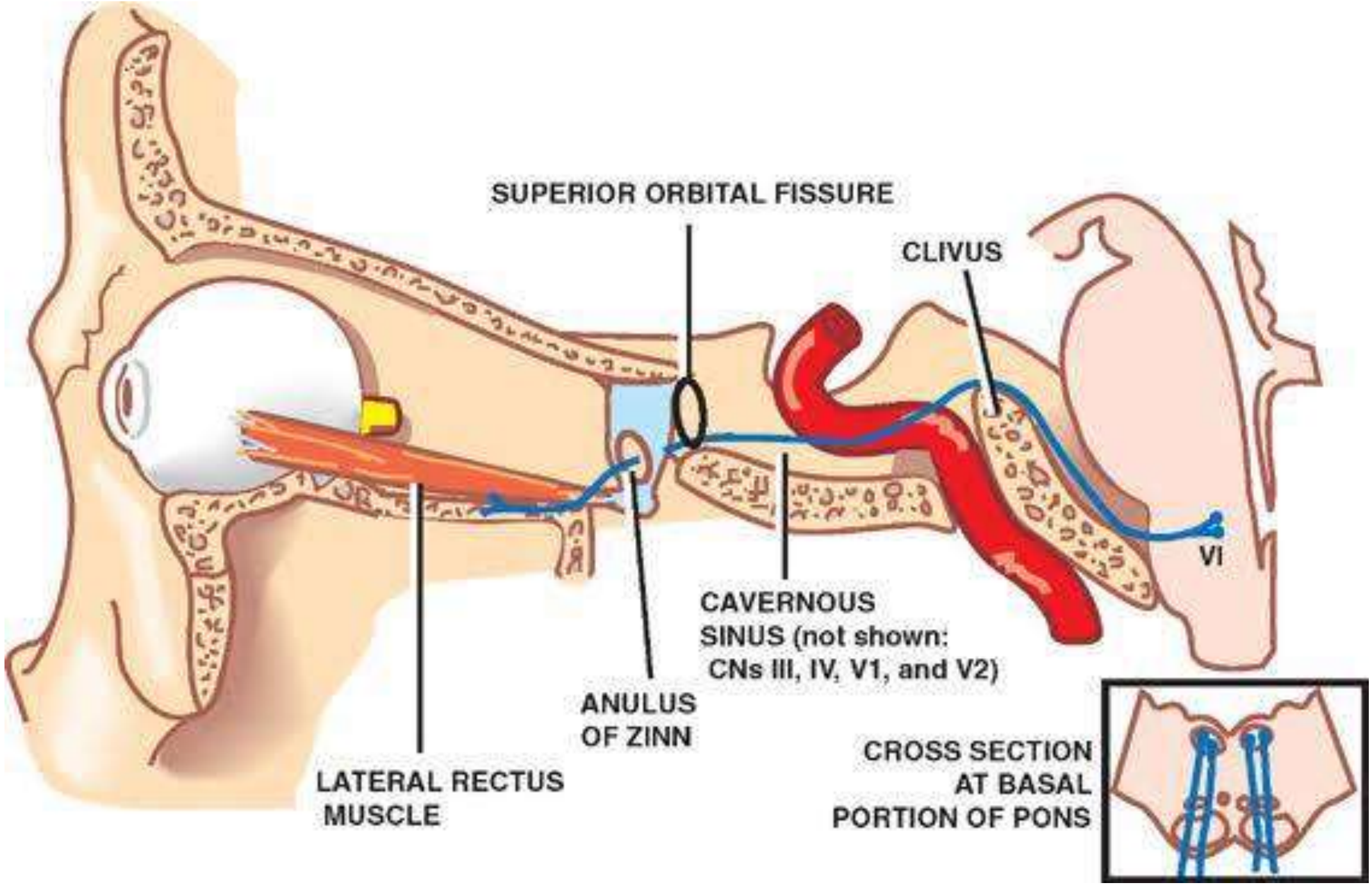


Course (cont’d)

- Nerve fibres travels anteriorly through pons in proximity to the pyramidal fibres.

- Nerve exits the pons at the pontomedullary junction to enter into the peri pontine cistern to cross the internal auditory artery.

- Nerve then climbs upward along the clivus of sphenoid bone near the basilar artery and Gasserian ganglion of the 5th nerve to make a sharp turn near the petrous bone apex.

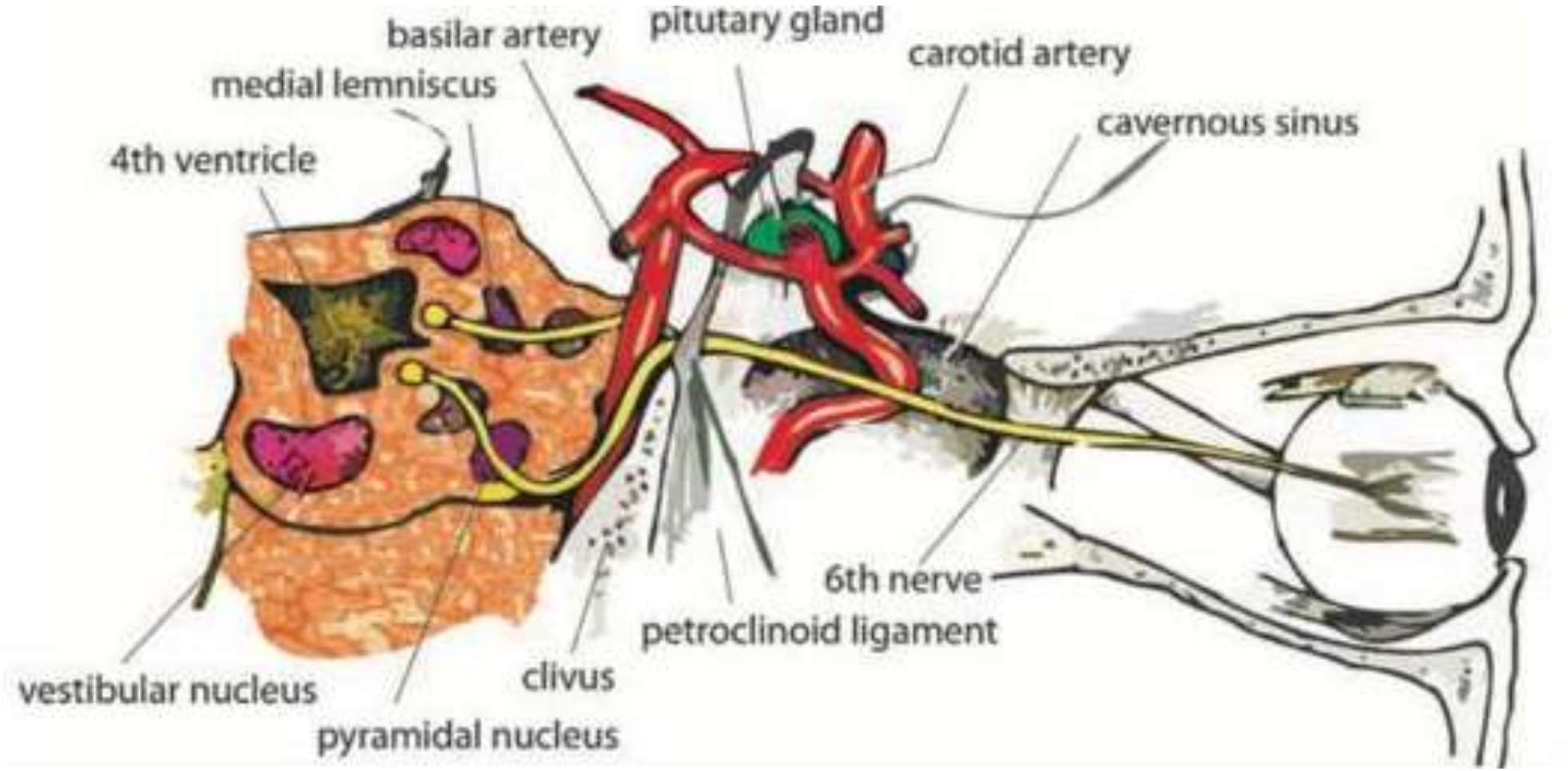

SUPERIOR ORBITAL FISSURE
CLIVUS
VI
CAVERNOUS
SINUS (not shown:
CNs III, IV, V1, and V2)
ANULUS
OF ZINN
LATERAL RECTUS
MUSCLE
CROSS SECTION
AT BASAL
PORTION OF PONS

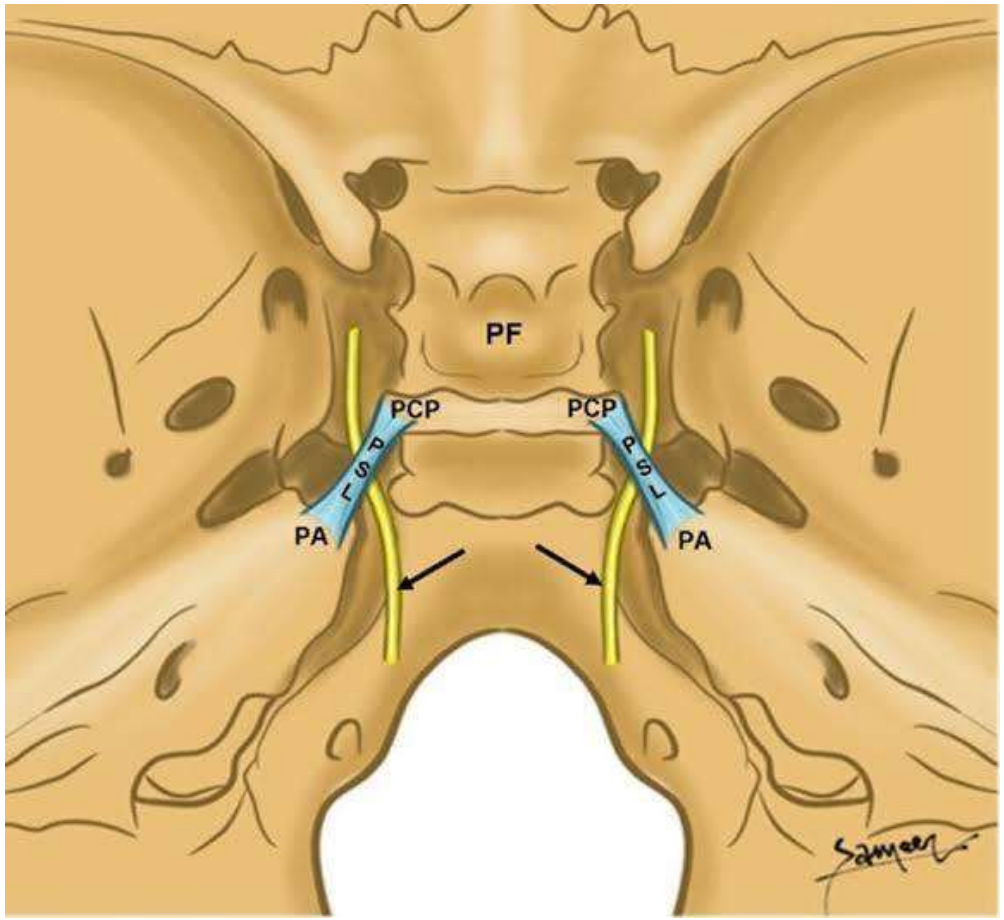
Course (cont’d)

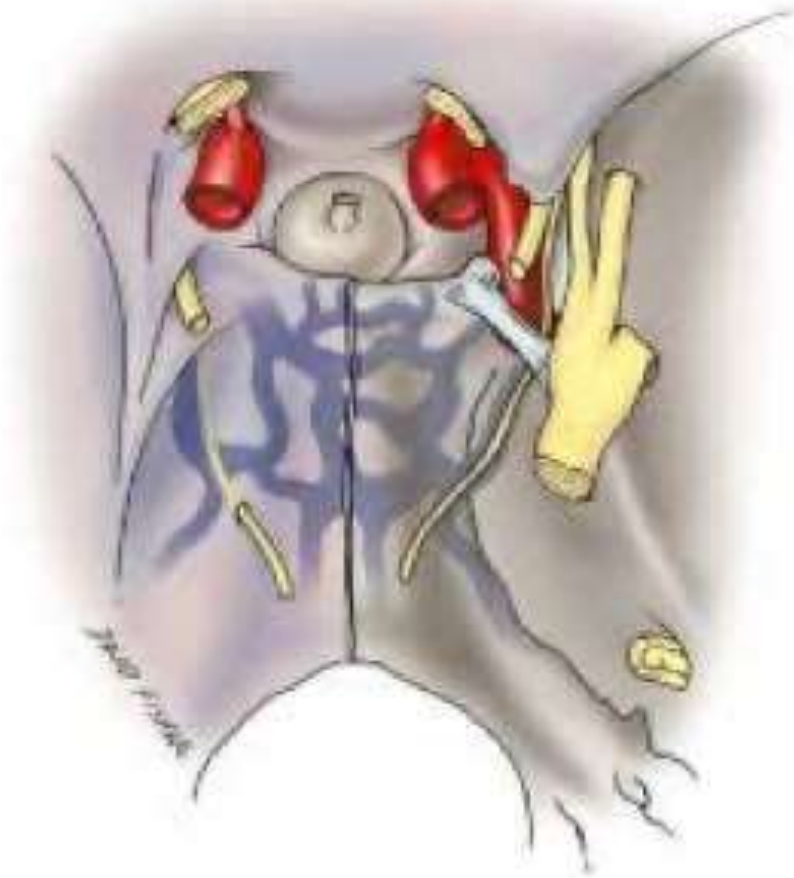
- Nerve then passes through the Dorello’s canal under the ligament of Gruber (petroclinoid ligament) and pierces the dura.

- The nerve then enters the cavernous sinus in close proximity to the ICA.

- The nerve then enters the orbit via the middle part of superior orbital fissure, to supply the lateral rectus muscle.

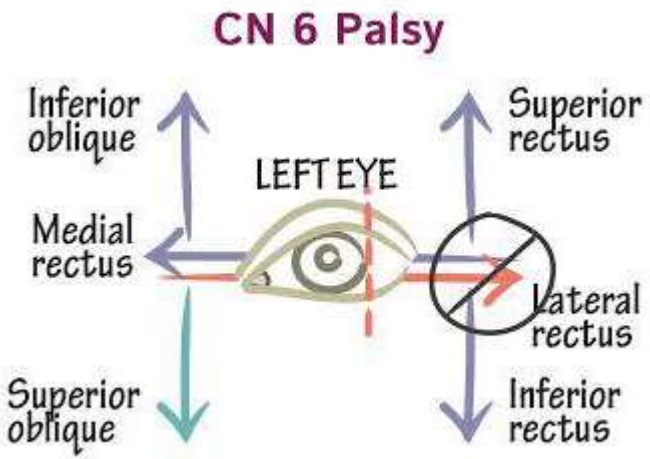

basilar artery
pitutary gland
carotid artery
medial lemniscus
cavernous sinus
4th ventricle
6th nerve
petroclinoid ligament
vestibular nucleus
clivus
pyramidal nucleus

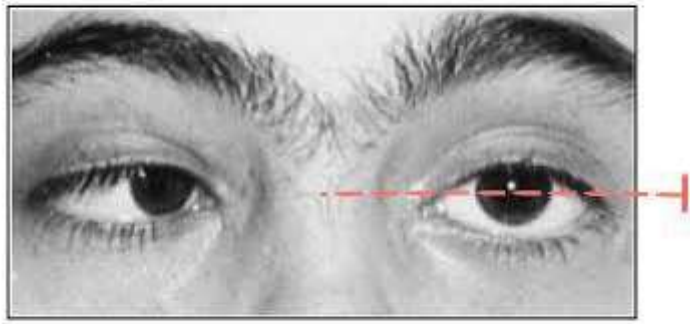

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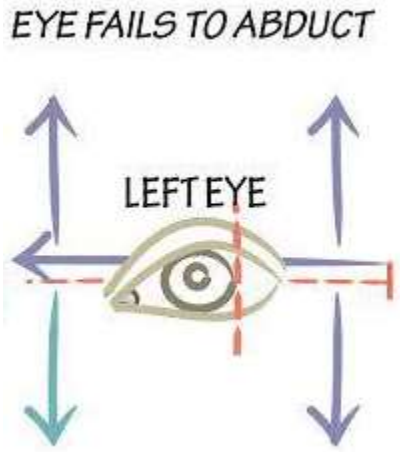


Function and Significance

- The nerve supplies the Lateral rectus muscle causing abduction of the eyeball.
- Palsy of the nerve causes horizontal gaze palsy, horizontal diplopia, more when looking towards the affected side and strabismus.







Primary and secondary gaze



Causes of 6th nerve palsies

- Trauma.
- Pontine stroke syndromes (Foville, Millard Gubler)
- Raised intracranial tension.
- Microvascular ischemia usually secondary to diabetes or hypertension.
- Infections or inflammation.
- Neoplasm or mass lesion compression. Metastasis as in metastatic prostate disease, Acoustic neuroma
- Vascular aneuryms especially basilar artery.
- Congenital disorders like congenital nuclear agenesis.
- Sarcoidosis.

Congenital nuclear agenesis

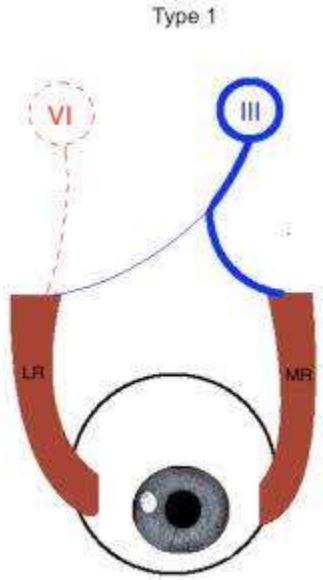
- Mobious Syndrome: 6th + 7th nerve nucleus agenesis (but can also present as other cranial nerve nucleus agenesis)

- Congenital cranial dysinnervation disorders (HOXA1 mutation: B/L Duane + Horizontal gaze palsy + progressive scoliosis)

Mobious Syndrome

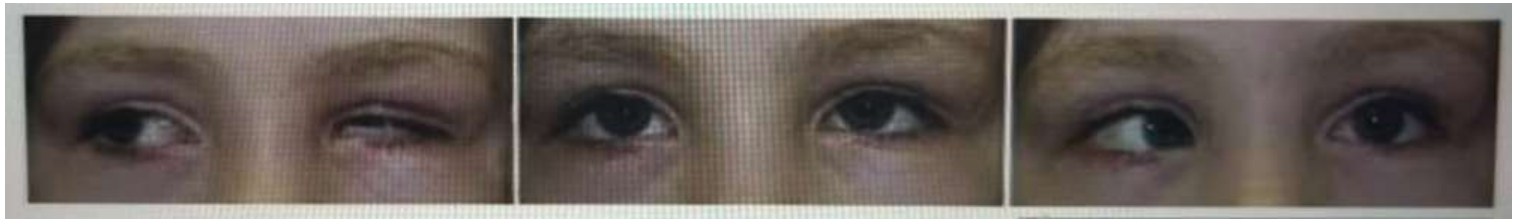
- Rhombencephalic maldevelopment.
- 6th nerve agenesis with facial diplegia.
- Duane retraction syndrome can occur.
- If 9th/ 10th cranial nerve involved, feeding and respiratory problems can occur

Duane Retraction Syndrome



- Type1:

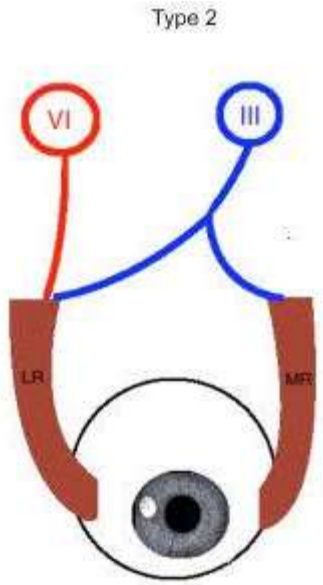
1. Agenesis of 6th nerve with cross innervation from 3rd nerve.
2. 3rd nerve supplies 75% to MR and 25% to LR
3. Abduction is limited but adduction is normal.
4. Eyeball retracts and palpebral fissure narrows on adduction of eyeball.



Duane retraction Syndrome

- Type2:

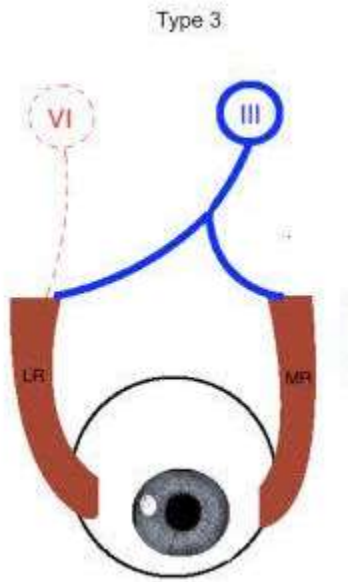
1. There is no agenesis of the 6th nerve nucleus, with equal innervation of MR and LR by the 3rd nerve. (150% innervation to LR and 50% innervation to MR)
2. Abduction is normal.
3. Adduction is impaired.



Duane Retraction Syndrome

- Type3

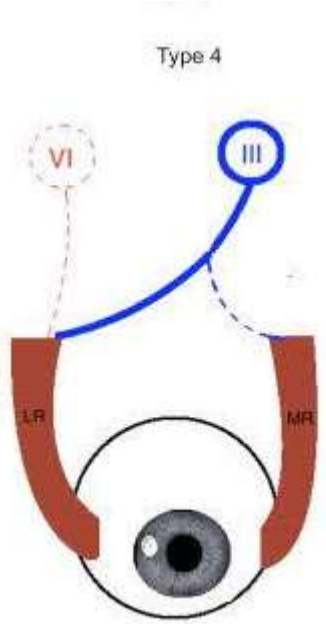
1. Agenesis of 6th nerve nucleus present, 3rd nerve equally supplies MR and LR
2. Abduction and Adduction both are impaired.



Duane Retraction Syndrome

- Type4:

1. Also called synergistic divergence.
2. Agenesis of 6th nerve nucleus present with 75% innervation of 3rd nerve to LR and 25% innervation to MR. (175% supply to LR and 25% to MR)
3. Paradoxical abduction on attempting to adduct the eye.



Lesions in Subarachnoid Space

- Gradenigo syndrome: due to infections/ mass lesions involving tip of petrous bone (like CSOM, Tumors, metastasis,etc) → Gasserian ganglion of 5th nerve, 6th nerve are involved.

1. Ipsilateral facial/ retro-orbital pain due to 5th nerve involvement.
2. 6th nerve palsy
3. Deafness due to proximity of 8th nerve to tip of petrous

- Godferson Syndrome: due to Carcinoma nasopharynx extending to clivus and foramen magnum causing 6th + 12th cranial nerve palsy.

Management

- Diagnostic modalities: Clinical examination, blood investigations to rule out systemic diseases, Imaging modalities like MRI/CT.

- Treatment strategies: Management of the underlying cause, prism lenses for diplopia, surgical procedures for strabismus.

References

- Harrison’s Textbook of Internal Medicine
- Bickerstaff’s Neurological Examination in Clinical Practice
- Bradley’s Neurology in Clinical Practice
- Localization in Clinical Neurology, Paul W. Brazis
- Up to Date

THANK YOU